Introduction to Module 7: Discrete Optimization

Design and Analysis of Algorithms

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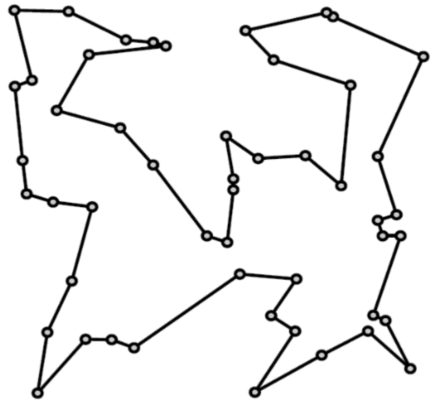
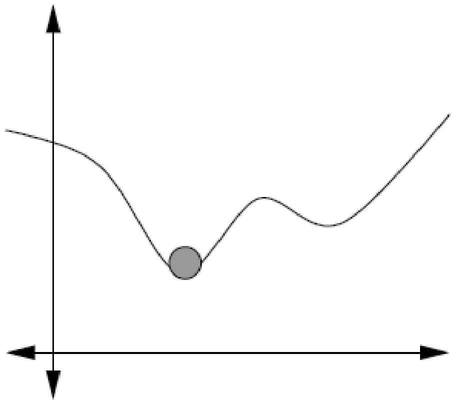
Optimization

- Find the best solution (i.e., of minimum or maximum value) from a set of alternatives.
- Solving and optimizing similar: for example, to minimize f(x), solve $f'(x) = 0$ (if f convex)
- Regression is the process of fitting parameters to a model to minimize its error when fitting to data.
- In this module, we focus on discrete optimization problems.

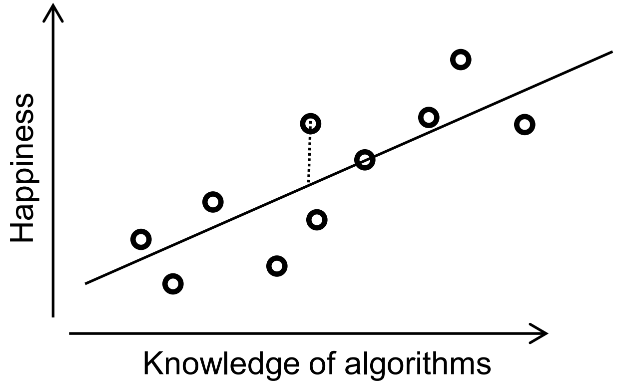
The traveling salesman problem (TSP)



Function minimization



Linear regression



Main Topics

- “Hard” vs “easy” problems
- Greedy algorithms
- Dynamic programming
- Heuristic methods for approaching hard problems (e.g., iterative refinement)
- Coding discussion: Exhaustive “Branch and bound” search for solving the traveling salesman problem.
- Extra topics: duality, relaxation
- Optimal enrichment lecture: Approximation algorithms

Programming Assignment

- Use heuristic methods (e.g., iterative refinement, greedy algorithms, relaxation and rounding, branch and bound search) to find good solutions to a hard optimization problem.
- Described in handout (not video); problem will rotate every semester.